Savannah Tank & Equipment Corp.

QUALITY CONTROL MANUAL

SHOP AND FIELD MANUFACTURE OF PRESSURE VESSELS IN ACCORDANCE WITH THE ASME BOILER & PRESSURE VESSEL CODE SECTION VIII DIV 1

AND

SHOP AND FIELD REPAIRS AND ALTERATIONS OF METALLIC PRESSURE-RETAINING ITEMS IN ACCORDANCE WITH THE NATIONAL BOARD INSPECTION CODE AND JURISDICTIONAL REQUIREMENTS

SAVANNAH TANK
AND EQUIPMENT CORPORATION
Tanks · Pressure Vessels · Process Equipment

☐ Controlled
☒ Uncontrolled 10th EDITION Manual number _____

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APPROVED BY: _____ _____
QUALITY CONTROL MANAGER DATE

ACCEPTED BY: _____ _____
AUTHORIZED INSPECTOR DATE

AUTHORITY AND RESPONSIBILITY

STATEMENT OF POLICY AND AUTHORITY

The Management of Savannah Tank & Equipment Corp. (ST & EC) is committed to the requirements outlined in this Quality Control Manual for the following scope of work:

- Shop and field manufacture of pressure vessels in accordance with ASME Code Section VIII Div 1.
- Shop and field repairs and alterations to metallic pressure-retaining items (PRI) including ASME Code Section I and Section IV Boilers, Section B31.1 Power Piping, Section VIII Div 1 Pressure Vessels and non-ASME code items in accordance with the National Board Inspection Code (NBIC), jurisdictional requirements and this Quality Control Manual.

As used throughout this Quality Control System the term "Code" shall include, as applicable, the original code of construction, ASME, NBIC and applicable jurisdictional rules.

The Quality Control Manager (QCM) has full support of the management and is responsible for the preparation, revision, approval, and issuance of the Quality Control Manual and the administration and implementation of the Quality Control System in the shop and field. The Quality Control Manager is assigned the responsibility, the authority, and the organizational freedom to identify Quality Control problems, stop work if necessary, initiate, recommend, and provide solutions.

Certifications include: authorization, approval, and certification. For ASME fabrication, certifications will only be by written signature/initial and written date unless otherwise described in the Quality Control Manual. The President of Savannah Tank & Equipment Corp. only allows those electronic signatures and approvals described in this Quality Control Manual. If other electronic methods are approved in the future, this Quality Control Manual will be revised to describe the controls and safeguards to be employed to ensure the integrity of the certification. Savannah Tank & Equipment Corp. utilizes the EDT System for certifying and registering Manufacturer's Data Reports.

When required by context in this Quality Control Manual, the feminine or masculine gender shall be treated as such other gender as appropriate.

Each individual is responsible for their assigned activities. They may delegate the performance of their duties to competent or qualified individuals as appropriate, but the responsibility shall not be delegated.

Any unresolved disagreement with the Quality Control Manager in regards to implementation of this Quality Control System shall be brought to me for resolution, and shall not conflict with code, jurisdiction/regulatory authority or Quality Control Manual requirements.

Signed: _____ Date: _____
PRESIDENT
Savannah Tank & Equipment Corp.

MANUAL CONTROL

This Quality Control Manual and its revisions shall be prepared and approved by the Quality Control Manager and submitted to the Authorized Inspector for review and acceptance prior to issuance and implementation. The approval of the Quality Control Manager and the acceptance of the Authorized Inspector (AI) are shown on the Table of Contents.

The Quality Control Manager shall review Code editions at the time of issue for possible changes that will require revisions to this Quality Control Manual. If revisions are required they shall be made within six (6) months of issue date. The Quality Control Manager shall document this review by generating a memo-to-file, or as an alternative he may initial and date the inside cover page of the new Code edition.

The Quality Control Manager shall process revisions by complete section with the revision level noted on the Table of Contents and in the footer of each section, except for the Exhibits section. The Exhibits section will be revised by individual exhibit with the revision level noted on the Exhibits section table of contents page and in the footer of each exhibit. The latest revised text will be indicated by ***bold italicized font***.

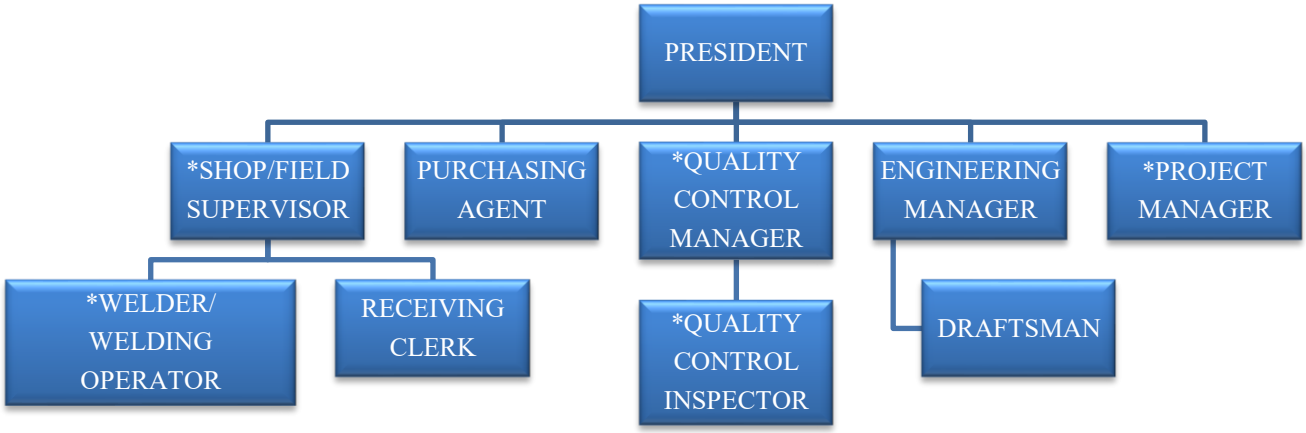
The Quality Control Manager will maintain and distribute all controlled copies of this Quality Control Manual and its revisions in accordance with a List of Manual Holders (Exhibit 1). Revisions will be inserted in the Quality Control Manual by the Quality Control Manager and the revised sections and pages shall be destroyed. Each controlled Quality Control Manual will bear the control number on its Title Page as is shown on the List of Manual Holders.

Uncontrolled Quality Control Manuals, as indicated on the Title Page, will not be issued to personnel within Savannah Tank & Equipment Corp., but may be issued to external parties upon request. These Quality Control Manuals will be current at time of issuance but will not be updated to include subsequent revisions.

A controlled copy of the Quality Control Manual is available to the Authorized Inspector at the shop or field site where Code activities are being carried out.

The Quality Control Manager, at his discretion, may promote this Quality Control Manual to the next numerical edition level (noted on the Title Page), at which time all previous revision levels will revert to zero and bold italicized font will be removed.

ORGANIZATION



* May be involved in both shop and field activities

DRAWINGS, DESIGN CALCULATIONS AND SPECIFICATION CONTROL

5.1 GENERAL

5.1.1 Calculations, drawings, and specifications may be generated in-house, subcontracted, or provided by the customer.

5.1.2 Each order for a pressure vessel to be completed by Savannah Tank & Equipment Corp. is assigned a job number by the Project Manager which is used to identify design and fabrication documents related to the pressure vessel. The job number also becomes the serial number of the vessel.

5.2 QUALIFICATION FOR ASME CODE SECTION VIII Div 1 DESIGN ACTIVITY

5.2.1 All appendix references listed in the 5.2 paragraphs below are as outlined in ASME Code Section VIII-1 Appendix 47.

5.2.2 The qualifications and experience required of the person in responsible charge of design activities will depend on the design complexity of the pressure vessel and the nature of the individual's experience. As such, more than one responsible charge may be designated.

5.2.3 The person in responsible charge shall be qualified to perform design work or exercise control of design work performed by others. The responsible charge shall be designated by the President. As outlined below, the responsible charge shall be either a Certifying Engineer, engineer, or a designer, and will be responsible for the design of the applicable pressure vessel from vessel conception to completion that is certified by Savannah Tank & Equipment Corp.

Savannah Tank & Equipment Corp. has designated the Engineering Manager as a responsible charge. Other personnel will be designated as applicable.

The qualification requirements shall also apply to Certifying Engineers, engineers and designers who are engaged by Savannah Tank & Equipment Corp. by contract or agreement for their services in the design of pressure vessels.

Certifying Engineer:
An engineer or other technical professional duly accredited and qualified to practice engineering activities as required by ASME Code Section VIII-1. A Certified Engineer shall be Chartered, Registered, or Licensed in accordance with Appendix 47-2(a)(1)(2).

Engineer:
The engineer shall have a degree from an accredited university or college in engineering, science, or technology requiring an equivalent of 4 years of full-time study of higher education, and 4 years or more of experience in the design of pressure vessels.

Designer:
The Designer shall meet either of the following:

- Completed an accredited engineering technician or associates degree, requiring the equivalent of at least 2 years of study, plus have a minimum of 6 years of experience in the design of pressure vessels.
- Have a minimum of 10 years of experience in the design of pressure vessels.

5.2.4 In lieu of the requirements listed in 5.2.3 above, Savannah Tank & Equipment Corp. may implement the following alternative qualifications for the engineer or designer in responsible charge:

Engineer:
The engineer shall have a degree from an accredited university or college in engineering, science, or technology requiring an equivalent of 4 years of full-time study of higher education, and a minimum of 1 years of experience in the design of pressure vessels.

Designer:
The designer shall meet either of the following:

- have completed an accredited engineering technician or associates degree, requiring the equivalent of at least 2 years of study, and a minimum of 1 year of experience.
  or
- have a minimum of 6 years of experience in the design of pressure vessels.

5.2.5 Individuals engaged in design activity while under the responsible charge shall:

- have knowledge of the design requirements of ASME Code Section VIII-1 for the application of the Certification Mark with the appropriate designator
- have knowledge and understanding of the requirements in this Quality Control Manual
- have training commensurate with the scope, complexity, criticality, or special nature of the design activities to which oversight is to be provided.

5.2.5.1 The Quality Control Manager shall maintain a documented record containing objective evidence of meeting the qualifications for the experience and training obtained.

5.2.5.2 Individuals engaged in design activity shall be permitted to engage in any design activity required by ASME Code Section VIII-1 as duly qualified.

5.2.6 Engineers and designers engaged in design activities which include numerical analysis, fatigue assessments, seismic reactions, quick-actuating closures, or U-2(g) design activities require additional qualifications. Appendix 47-5 / Table 47-5-1 shall be adhered to in such instances.

5.2.7 The Quality Control Manager shall maintain the Design Continuity Log (Exhibit 10) to record annual proficiency of each Certifying Engineer/engineer/designer. This log shall be maintained by recording evidence of design activity within 1 year of designation, or by taking a design course that focuses on design of ASME Code Section VIII-1 pressure vessels.

5.3 DESIGN BY SAVANNAH TANK & EQUIPMENT CORP.

5.3.1 Upon receipt of an order for a pressure vessel to be designed by Savannah Tank & Equipment Corp., the customer's specification is reviewed for adequacy by the Engineering Manager and for compliance to Code requirements by the Quality Control Manager. Any deficiencies are reconciled with the customer prior to starting any design work. The Engineering Manager is responsible for submitting drawings to the customer for approval if required. Upon acceptance of the customer's specification, the Engineering Manager shall determine the design, material, nondestructive examination requirements, and all other Code criteria.

5.3.2 The Engineering Manager will prepare the drawings, calculations, and material specifications. At the option of the Engineering Manager a draftsman can be used to prepare drawings. All drawings shall address sufficient information to satisfactorily perform fabrication of the pressure vessel. Material specifications are listed on the drawing or as an alternative may be listed on a separate Bill of Material (Exhibit 2).

5.3.3 Design functions may be subcontracted. If subcontracted, the Engineering Manager will review the design documents for technical content, and the Quality Control Manager will review them for compliance to the Code. Any discrepancies are brought to the attention of the originator of the design for correction. Savannah Tank & Equipment Corp. takes full responsibility for all subcontracted design.

5.3.4 Completed design documents as listed above, are submitted to the Quality Control Manager for review. When the design meets Code requirements, the Quality Control Manager will approve each document.

5.3.5 Computer programs may be used to generate the design calculations provided the Engineering Manager validates the computer program capability by comparison to alternate methods prior to first use and upon each revision to the computer program and all new editions of the code of construction. Verification against examples found in ASME PTB-4, ASME Section VIII Div. 1 Example Problem Manual may be sufficient to show the verification is met. In lieu of above, the computer programs may be verified by the results confirmed by design analysis for each application. This shall be documented by maintaining the results of the chosen method and made available to the Authorized Inspector for review upon request

5.4 DESIGN BY OTHERS

5.4.3 When design documents are provided to Savannah Tank & Equipment Corp. by others, they are reviewed for adequacy by the Engineering Manager. The Quality Control Manager will review them for compliance to the Code. Any discrepancies are brought to the attention of the originator of the design for correction. Upon acceptance, the Engineering Manager shall determine the design, material, nondestructive examination requirements, and all other Code criteria.

5.4.4 Savannah Tank & Equipment Corp. takes full responsibility for the design.

5.4.5 When the design meets Code requirements, the Engineering Manager and the Quality Control Manager will approve the design documents.

5.5 DISTRIBUTION

5.5.3 The Engineering Manager releases the job to the shop or field by distribution of the applicable design and fabrication documents to the Shop/Field Supervisor and discussing the scope of work. The Shop/Field Supervisor will then distribute the documents to the Welder/Welding Operator and discuss the scope of work and fabrication requirements.

5.5.4 The Quality Control Manager is responsible for providing the Authorized Inspector access to all design documents for review prior to the start of work.

5.5.5 Revisions to design documents are processed in the same manner as the originals.

5.5.6 Superseded drawings are retrieved and destroyed by the Engineering Manager, unless they contain notes which require their retention, in which case the Engineering Manager will mark the documents "VOID" or some other marking that clearly indicates that they have been superseded.

5.6 RETENTION OF DESIGN DOCUMENTS

5.6.3 Design documents are maintained in accordance with Section 13 of this Quality Control Manual.

MATERIAL CONTROL

6.1 MATERIAL REQUISITION

6.1.1 All Purchase Requisitions (Exhibit 3A), either electronic in the Job Boss order system or paper copy, for shop and field Code material are prepared from approved material specifications listed on the drawing or a separate Bill of Material as applicable by the Project Manager .

6.1.2 Weld material may be ordered as stock without the use of a Purchase Requisition and may not be necessarily shown on the drawing or on a Bill of Material.

6.1.3 The Purchase Requisition shall list materials selected from the appropriate stress table referenced in the applicable code of construction, by the SA, SB, or SFA specification. The applicable material grade, type or alloy, and any special testing or material test report requirements, as determined by the Project Manager, shall be listed as well. Examples are: UG-79 requirements, impact tests, etc.

6.1.4 Completed Purchase Requisitions are reviewed for Code compliance by the Project Manager. When acceptable he will approve the Purchase Requisition electronically in the password protected Job Boss system or initial and date on the paper copy of the document.

6.1.5 Material substitutions are not allowed unless they are approved by the Engineering Manager, the Quality Control Manager and concurrence is obtained from the Authorized Inspector. The owner's concurrence must be obtained when required.

6.1.6 Purchase Requisitions for material other than weld material by field personnel will not be allowed.

6.2 PURCHASE ORDERS

6.2.1 The Project Manager electronically forwards the approved Purchase Requisition in the Job Boss system to the Purchasing Agent who will generate a Purchase Order (Exhibit 3B) from the information on the Purchase Requisition. All information will be transferred to the Purchase Order. The Purchase Order will then be distributed to the selected vendor.

6.2.2 A copy of the Purchase Order is maintained on the P-Drive and accessed by the Quality Control Inspector or the Receiving Clerk for use during receiving inspection.

6.2.3 Revisions to Purchase Orders are processed in the same manner as the originals.

6.3 RECEIVING INSPECTION

6.3.1 All Code material received at the shop or field site is placed into a hold area pending receiving inspection.

6.3.2 The Quality Control Inspector or Receiving Clerk, using a copy of the Purchase Order for the P-Drive examines all Code material for condition, quantity, and identification marking to provide acceptability to the Purchase Order.

6.3.3 A Receiving Inspection Report (Exhibit 4) is completed for all Code material. The Quality Control Inspector or Receiving Clerk will record the dimensional findings and all existing material identification markings on the Receiving Inspection Report. Upon completion he will initial and date and forward it to the Quality Control Manager.

6.3.4 Weld material may be received as stock, and may be received without the use of a Receiving Inspection Report. If this method is chosen, the Quality Control Inspector will verify proper material marking on the container against the applicable specification requirements listed in ASME Code Section II Part C. If acceptable, he will place the weld material into the code weld storage. The act of placing the weld material into storage releases it for production.

6.3.5 For product forms which the code of construction requires material test reports, the Quality Control Manager reviews the material test report against the requirements listed in ASME Code Section II. If the material test report is acceptable, he will approve the material test report.

6.3.6 The Quality Control Manager shall verify the original product marking against the material specification or material standard as applicable.

6.3.7 If the dimensions, the product marking, and if applicable, the material test report meets Code requirements the Quality Control Manager will approve the Receiving Inspection Report.

6.3.8 Upon approval of the Receiving Inspection Report, the Quality Control Inspector will mark the material with the Purchase Order number and heat/slab number and either place the material in stock, or if ordered for a job he will also mark the material with the job number and release the material for production.

6.3.9 Small items that are marked with the required original material identification may be placed in bags, boxes, or bins until assigned to a job in lieu of marking them with a Purchase Order number. Prior to releasing to a job, the Quality Control Inspector shall mark these items with the job number. If the item is too small, the job number shall be identified on the container.

6.3.10 Code stamped parts are unloaded in the assigned hold area and inspected by the Quality Control Inspector for damage, applicable Code stamping and identification marking to provide traceability to the Manufacturer's Data Reports and/or assembly drawings. The applicable Manufacturer's Data Report must be received and verified against the part stamping before the item can be released by the Quality Control Inspector for production. This verification shall be documented on a Receiving Inspection Report.

6.3.11 The Quality Control Manager shall ensure that tests are performed and documented if further material testing is required to be performed at receiving inspection or during the manufacturing operations.

6.3.12 Material that does not meet all Purchase Order and Code requirements is retained in the hold area and controlled as described in Section 8 of this Quality Control Manual.

6.4 PRESSURE PART MATERIAL TRACEABILITY

6.4.1 Original pressure part material identification shall be maintained and plainly visible on the completed vessel. Prior to obliteration of the original material identification or when cutting Code material into two or more pieces, the original material identification is transferred to each piece by the Welder/Welding Operator. As an alternative, the required markings may be recorded and transferred using methods such as material tabulations or as-built sketches that ensure identification of each piece of material during fabrication and subsequent identification in the completed vessel. Another alternative, may be by applying a coded marking, which is traceable to the original required marking then transferred by the Welder/Welding Operator. The coded marking shall consist of the Purchase Order number and the material heat/slab number. Such transfers of markings shall be made prior to cutting. The method of marking shall be acceptable to the Authorized Inspector.

6.4.2 As an alternative to the coded marking requirements listed above, a Material Traceability Log (Exhibit 4A) may be utilized for maintaining traceability.

6.4.3 The maintenance of identification for welded non-pressure parts, such as skirts, supports, baffles, lugs, clips, etc. shall be at least to material type or material specification or P number.

6.5 STOCK MATERIAL

6.5.1 Stock material that is not identified with the Purchase Order number shall be received as described above using procurement documents if available. Stock material that is identified with a Purchase Order number will be verified to meet the current Code requirements prior to use. Upon acceptance, the Quality Control Inspector shall mark the material with the job number and release the material for production.

6.6 CUSTOMER SUPPLIED MATERIAL

6.6.1 All material supplied by the customer is received and inspected as described in paragraph 6.3 above. A Bill of Material and a Receiving Inspection Report are used during receiving inspection to assure Code compliance. Upon acceptance, the Quality Control Inspector shall mark the material with the job number and release the material for production.

6.6.2 Savannah Tank & Equipment Corp. retains Code responsibility for customer supplied material.

6.7 MATERIAL FORMED BY VENDORS

6.7.1 When material is formed into shapes by our vendors and the original markings as required by the applicable material specification are unavoidably cut out, or the material is divided into two or more parts, the vendor shall either transfer the original identification markings to another location on the shape; or provide for identification using a coded marking system

previously agreed upon as required on our Purchase Order. Savannah Tank & Equipment Corp. will accept either the supplier coded marking system or complete material identification marking.

6.7.2 The Quality Control Manager is responsible for verifying that the vendor has verified the original markings prior to processing.

6.8 RETENTION OF MATERIAL RECORDS

6.8.1 Material control documents are maintained in accordance with Section 13 of this Quality Control Manual.

EXAMINATION AND INSPECTION PROGRAM

7.1 SHOP FABRICATION

7.1.1 The Quality Control Manager shall provide for the Authorized Inspector to have access to all drawings, calculations, specifications, procedures, Traveler (Exhibit 5) and Weld Map (Exhibit 6), repair procedures, Proof Test Reports, records, test results, and any other documents as necessary for him to perform his duties.

7.1.2 Fabrication of pressure vessels is controlled by means of the Traveler, prepared by the Quality Control Manager. The Traveler contains the various document reviews, the stages of production, inspection, examination, and testing requirements. The Traveler also contains space for the initials and date of the Quality Control Inspector and the Authorized Inspector for the steps that are complete and acceptable.

7.1.3 When completed the Quality Control Manager reviews the Traveler and the drawings with the Authorized Inspector for designation of hold points. At a minimum, hold points shall be established to assure that all Code required inspections by Savannah Tank & Equipment Corp. and the Authorized Inspector are performed. The Quality Control Manager and the Authorized Inspector will initial and date the Traveler to document their review prior to the start of fabrication.

7.1.4 When a vessel is not provided with an inspection opening permitting access to inspect internal surfaces, a hold point shall be assigned for the Quality Control Inspector and the Authorized Inspector to permit internal inspection before making the final closure weld.

7.1.5 The Quality Control Manager shall apprise the Authorized Inspector of work progress, and give notification in advance of approaching hold points as designated on the Traveler. Work shall not proceed beyond the Authorized Inspector's hold points until acceptance of the operation is denoted on the Traveler.

7.2 FIELD FABRICATION

7.2.1 Field fabrication is controlled as described above except that the Quality Control Inspector carries out those functions designated in the shop for the Quality Control Manager.

7.3 PRESSURE TEST

7.3.1 The final pressure test will be applied and witnessed as required by the Code by the QC Inspector. The test may then be witnessed by the Authorized Inspector. The required test pressure will be indicated on the Traveler.

7.3.2 An indicating gage shall be connected directly to the vessel or with a pressure line that does not include intermediate valves. Hydrostatic head on the gage shall be considered such that the required test pressure is achieved at every point in the vessel. If a dial indicating pressure gage is used, the range shall be about double the intended maximum test pressure but in no case less than one and one half (1-1/2) nor more than four (4) times that pressure. Digital reading pressure gages having a wider range of pressure may be used provided the readings give the same or greater degree of accuracy as obtained with the dial pressure gages. The spacing between graduations shall be such that the Authorized Inspector and the operator controlling the test shall be able to determine when the required test pressure has been applied.

7.3.3 If the indicating gage is not readily visible to the operator controlling the pressure applied, an additional indicating gage shall be provided where it will be visible to the operator throughout the duration of the test.

7.4 FINAL INSPECTION

7.4.1 When the fabrication or assembly of a pressure vessel is completed, the Quality Control Inspector and the Authorized Inspector will make their final inspection.

7.5 DATA REPORTS AND STAMPING

7.5.1 To prevent loss or unauthorized use, the Quality Control Manager shall have custody and control of the ASME Certification Mark and the 'NB' registration stamp and shall issue them to the Quality Control Inspector in the shop or at the field site as needed. The ASME Certification Stamp may not be loaned out or used by anyone other than Savannah Tank & Equipment Corp.

7.5.2 The ASME Certification Stamp shall be returned to the ASME upon demand or if Savannah Tank & Equipment Corp. discontinues the use or if no inspection agreement exist with an AIA or if the Certificate of Authorization has expired a new certificate will not be issued.

7.5.3 When satisfied that all Code requirements have been met, the Authorized Inspector will verify the stamping or nameplate data and authorize and verify the application of the ASME Certification Mark. The ASME Certification Mark, the 'NB' registration stamp and all other required nameplate stamping shall be stamped on the pressure vessel or on a nameplate attached to the item. The Quality Control Inspector shall apply the stamping or the attachment of the nameplate to the item.

7.5.4 The abbreviation 'ST & EC' may be stamped on the pressure vessel or the nameplate as appropriate, in lieu of Savannah Tank & Equipment Corp..

7.5.5 The Quality Control Manager will prepare and certify the Manufacturer's Data Report upon completion of a fabricated pressure vessel.

7.5.6 The Quality Control Manager will complete and certify the field assembly certification block on the applicable Manufacturer's Data Report upon completion of the field assembly of a pressure vessel.

7.5.7 The completed Manufacturer's Data Report and final records will be presented to the Authorized Inspector for review and certification.

7.5.8 The Quality Control Manager shall furnish a copy of the Manufacturer's Data Report, after certification, to the user and, upon request to the Authorized Inspector, and if applicable to the appropriate enforcement authority in the jurisdiction in which the vessel is to be installed. The Quality Control Manager shall either keep a copy of the Manufacturer's Data Report on file in a safe repository for at least three years or he will register the vessel and file the original Data Report with the National Board.

7.5.9 The Quality Control Manager shall forward Partial Data Reports in duplicate to the manufacturer of the pressure vessel. When used to record parts furnished to the user of an existing pressure vessel as replacement or repair parts, the Partial Data Report is sufficient and need not support a Master Data Report. If this is the case, the Partial Data Report shall be forwarded to the user.

7.5.10 If Manufacturer's Data Reports are registered with the National Board, the National Board registration numbers shall be controlled and assigned by the Quality Control Manager following the pressure test. The National Board registration number shall be assigned without skips or gaps of unused numbers or duplication of numbers. The Quality Control Manager shall submit the original Manufacturer's Data Report to the National Board within 30 days following certification. The Quality Control Manager shall maintain a National Board log showing the National Board registration number, the date certified, the date sent to the National Board, and the manufacturer's serial number. The National Board Log is not required to be maintained if registering Manufacturer's Data Reports via the EDT System.

7.6 NONCONFORMITIES

7.6.1 All nonconformities found during fabrication, installation, or assembly are controlled as described in Section 8 of this Quality Control Manual.

7.7 RECORD RETENTION

7.7.1 All records referenced in this section are maintained in accordance with Section 13 of this Quality Control Manual.

CORRECTION OF NONCONFORMITIES

8.1 IDENTIFICATION OF NONCONFORMITIES

8.1.1 A nonconformity is any condition that does not comply with the applicable rules of the Code, the requirements of this Quality Control Manual, or other specified requirements. All nonconformities shall be corrected or eliminated prior to being released from the nonconforming status.

8.1.2 It is the duty of all Savannah Tank & Equipment Corp. employees to report nonconformities to their supervisor who shall notify the Quality Control Manager or Quality Control Inspector. The Quality Control Inspector shall verify the nonconforming condition, and if applicable will tag the item with a hold tag or mark the item 'hold'. The Quality Control Inspector will generate a Non-conformance Report (NCR) (Exhibit 8). If practical, the nonconforming item shall be moved to a segregated hold area.

8.2 DISPOSITIONS

8.2.1 USE-AS-IS: When the disposition is use-as-is the Quality Control Manager will consult with and obtain the approval of the Engineering Manager and concurrence of the Authorized Inspector. Any required revisions to drawings, calculations, and procurement documents will be controlled as described in this Quality Control Manual.

8.2.2 REPAIR TO MATERIAL: When the disposition requires a defect repair, the method and extent of the repair shall be approved by the Quality Control Manager and accepted by the Authorized Inspector. The method and extent of repair shall be documented on the Nonconformance Report.

8.2.3 RETURN TO VENDOR: When the disposition is return to vendor, the Quality Control Inspector shall verify that the item has been removed from the work area, if possible, and is clearly marked to prevent its inadvertent use prior to return.

8.2.4 OTHER: The Quality Control Inspector shall present all nonconforming dispositions that don't fit into a category listed above to the Quality Control Manager for review and approval. If applicable, the Quality Control Manager shall obtain concurrence of the Authorized Inspector.

8.3 CORRECTION OF NONCONFORMITIES

8.3.1 The Non-conformance Report, together with the suggested disposition signed by the Quality Control Inspector is forwarded to the Quality Control Manager for review. The Engineering Manager will provide technical assistance and approval if applicable.

8.3.2 Upon correction, the Quality Control Inspector shall reinspect the item as dispositioned above. When Code requirements and the dispositions have been met he will forward the Non-conformance Reports to the Quality Control Manager for final close-out, and if applicable to the Authorized Inspector for concurrence. After the non-conformance has been resolved the Quality Control Inspector will remove the hold tag or marking and release the item from the nonconforming status.

8.3.3 As applicable, the Quality Control Inspector will record the NCR number either on the Traveler or the Receiving Inspection Report.

8.4 RECORD RETENTION

8.4.1 Non-conformance Reports are maintained in accordance with Section 13 of this Quality Control Manual.

WELDING CONTROL

9.1 GENERAL

9.1.1 All welding shall be performed using Welding Procedure Specifications (WPS) and Welders or Welding Operators qualified in accordance with the requirements of ASME Code Section IX and the applicable code of construction.

9.1.2 Personnel that are assigned with responsibility for supervision, control, evaluation, and acceptance of qualification testing shall be designated by the President and/or the Quality Control Manager. The designated individual shall have a satisfactory level of competence. As a minimum, this individual shall be qualified by education, experience, or training in the following areas:

a. knowledge of the requirements of ASME Code Section IX for the qualification of procedures and/or joining personnel
b. knowledge of this Quality Control Manual
c. knowledge of the scope, complexity, or special nature of the activities to which oversight is to be provided.

For a candidate to qualify for this position, a minimum of 1 year welding supervisory experience is required to assess sufficient knowledge of welding technology and Code requirements.

Welding supervisory duties as described above, shall not be delegated to a non-qualified individual.

The Quality Control Manager shall maintain a record containing objective evidence of the qualifications, training, or experience.

9.2 WELDING PROCEDURE SPECIFICATION

9.2.1 Welding Procedure Specifications are written by the Quality Control Manager and test welds are made under the Quality Control Manager's full supervision and control. Preparation and testing of the required specimens is performed to the requirements of ASME Code Section IX by a testing laboratory. The recorded variables, and the test report results are reviewed by the Quality Control Manager. If acceptable, he will prepare and certify the Procedure Qualification Record (PQR).

9.2.2 The Quality Control Manager will ensure that Welding Procedure Specifications are available to the Welders and Welding Operators in the work area.

9.2.3 Welding Procedure Specifications are revised when there is a change in a nonessential variable and requalified when there is a change in an essential or required supplementary essential variable.

9.2.4 Welding Procedure Specification records and Procedure Qualification Records are submitted to the Authorized Inspector for review. The Authorized Inspector has the right at any time to call for and witness the tests of the Welding Procedure Specification.

9.2.5 Standard Welding Procedure Specifications (SWPS) listed in ASME Code Section IX may be used as an alternative, provided the SWPS is demonstrated and qualified by welding and testing one groove weld test coupon following that SWPS in accordance with ASME Code Section IX paragraph QW-500. The Quality Control Manager shall accept by certification full responsibility for the application of the SWPS. When using Standard Welding Procedure Specifications all variables listed are considered essential and, therefore, deviations, modifications, amendments, or revisions are prohibited.

9.3 QUALIFICATION OF WELDERS AND WELDING OPERATORS

9.3.1 All Welders and Welding Operators to be used for Code welding are qualified to ASME Code Section IX under the full supervision and control of the Quality Control Manager. The preparation and testing of the required specimens is performed in-house or by a testing laboratory. The recorded variables, and the test report results are reviewed by the Quality Control Manager. If acceptable, he will prepare and certify the Welder/Welding Operator Performance Qualification (WPQ/WOPQ) record.

9.3.2 The Quality Control Manager shall provide copies of the WPQ/WOPQ records to the Shop/Field Supervisor for their respective Welders/Welding Operators in the work area.

9.3.3 The Quality Control Manager shall make the WPQ/WOPQ records available to the Authorized Inspector for review. The Authorized Inspector has the right at any time to call for and witness tests of the welding procedure or of the ability of any welder and welding operator.

9.3.4 A Welder/Welding Operator is required to be requalified when a change is made in one or more essential variables for that process.

9.4 EXPIRATION AND REVOCATION OF QUALIFICATION

a. CONTINUITY: The performance qualification of a welder or welding operator shall remain valid, provided no greater than 6 months have passed since the qualified welding process was last used under the supervision and control of the qualifying or participating organization(s), by the

   (1) Welder using manual or semiautomatic welding, or

   (2) Welding Operator using a machine or automatic welding

b. REVOCATION: When there is a specific reason to question the Welder's or Welding Operator's ability to make welds that meet the specification, the qualifications that support the welding that is being performed shall be revoked. All other qualifications not questioned will remain in effect.

9.5 MAINTENANCE OF WELDER WELDING OPERATOR QUALIFICATIONS

9.5.1 Each Welder and Welding Operator is listed on the Welders Log (Exhibit 7) maintained by the Quality Control Manager from data provided by the Shop/Field Supervisor. The Shop/Field Supervisor determines from the log when a Welder or Welding Operator's qualifications are about to expire so he can assure that the Welder or Welding Operator welds with the required process or is properly requalified.

9.6 PRODUCTION WELDING

9.6.1 The Shop/Field Supervisor is responsible for verifying that all production Welders/Welding Operators are properly qualified and for instructing them in the Welding Procedure Specification they are to use as listed on the drawing.

9.6.2 Each qualified Welder/Welding Operator is issued a unique identification by the Quality Control Manager. The Welder/Welding Operator may stamp the identification on or adjacent to and at intervals of not more than 36" along the welds which he makes. The limitation of stamping of a steel plate shall be ¼" and over in thickness and in nonferrous plates ½" and over in thickness. However, the Quality Control Manager shall enter the Welder/Welding Operator identification on the Weld Map for each weld made.

9.6.3 Tack welds whether left in place or removed are made by using a qualified Welding Procedure Specification.

9.6.4 Tack welds left in place are made by qualified Welders using a qualified Welding Procedure Specification.

9.6.5 If a Welder making the tack weld that is left in place, is not the same Welder that finishes the weld, that Welder shall identify each tacked joint as described in paragraph 9.6.2 above.

9.6.6 Tack welds left in place shall be properly prepared for inclusion into the weld, visually examined by the Welder, and if found defective shall be removed.

9.6.7 Tack welds made by subcontractors shall be removed.

9.7 WELDING MATERIAL

9.7.1 Welding material is visually inspected by the Welder/Welding Operator prior to use to ensure that it is flagged or marked with the AWS classification in accordance with the Welding Procedure Specification.

9.7.2 All welding materials are stored in a dry storage area and issued to the shop or field work areas by the Shop/Field Supervisor.

9.8 RECORD RETENTION

9.8.1 All welding documents are maintained in accordance with Section 13 of this Quality Control Manual.

NONDESTRUCTIVE EXAMINATION

10.1 GENERAL

10.1.1 All Nondestructive Examination (NDE) required by the Code may be performed by NDE subcontractors, whose written practice and personnel qualifications have been reviewed and accepted by the Quality Control Manager.

10.1.2 Savannah Tank & Equipment Corp. may choose to perform some liquid penetrant (PT) examinations. These examinations will be performed using procedures, approved by the Quality Control Manager, and utilizing properly certified in-house personnel.

10.2 NDE PROCEDURES

10.2.1 All procedures are detailed enough to provide the technique to meet ASME Code Section V requirements and they are accepted for use by the Quality Control Manager.

10.2.2 All NDE procedures shall be demonstrated to the satisfaction of the Authorized Inspector.

10.2.3 All Subcontracted NDE is performed using written procedures that have been prepared and accepted by a Level III examiner certified in the specific method.

10.3 SUBCONTRACTED NDE PERSONNEL

10.3.1 Subcontracted personnel shall be qualified and certified in accordance with their employer's written practice which is written to meet the requirements of the latest Code accepted edition of the SNT-TC-1A or CP-189. (as amended by ASME Code Section V Article 1 T-120 Mandatory Appendix III). The Quality Control Manager will review and accept the qualification records of the subcontracted Level III examiner to assure that he has the training and experience and is certified for the methods in accordance with the written practice. The Quality Control Manager will also review and accept the certification record of each Level I and II examiner prior to performing any examination to ensure that they are certified by the Level III examiner.

10.3.2 In addition the requirements described above, if the techniques of computed radiography, digital radiography, phased-array ultrasonic, or ultrasonic time-of-flight diffraction are to be used, the training, experience, and examination requirements found in ASME Code Section V Article 1, Mandatory Appendix II shall also be included in the employer's written practice for each technique as applicable.

10.3.3 Subcontracted NDE personnel qualification records are made available to the Authorized Inspector for review.

10.4 NDE REPORTS

10.4.1 All radiographs and reports of NDE are reviewed and accepted, as applicable, by the Quality Control Manager and maintained in accordance with Section 13 of this Quality Control Manual. All radiographs and reports of NDE are made available to the Authorized Inspector.

10.5 QUALIFICATION OF SAVANNAH TANK & EQUIPMENT CORP. NDE PERSONNEL

10.5.1 Examiners who perform in-house liquid penetrant examinations shall be qualified and certified by the Quality Control Manager in accordance with ASME Code Section VIII-1 Appendix 8 which requires the following:

- Has vision, with correction if necessary, to enable him to read a Jaeger Type No. 2 Standard Chart at a distance of not less than 12 in., and is capable of distinguishing and differentiating contrast between colors used. These requirements shall be checked annually.

- Is competent in the techniques of the liquid penetrant examination method for which he is certified, including making the examination and interpreting and evaluating the results, except that, where the examination method consists of more than one operation, he may be certified as being qualified only for one or more of these operations.

10.5.2 Upon completion the Quality Control Manager shall certify each examiner to be competent.

10.5.3 The following criteria may be used as an alternate to the above certification requirements:

Examiners qualified to a written practice in accordance with the latest edition of the following standards:

- SNT-TC-1A – the Recommended Practice for Nondestructive Testing Personnel Qualification and Certification.
- CP-189 – the ASNT Standard for Qualification and Certification of Nondestructive Testing Personnel.

10.5.4 In-house NDE personnel qualification records are made available to the Authorized Inspector for review.

HEAT TREATMENT

11.1 GENERAL

11.1.1 All heat treatment, such as, pre-heat; annealing; normalizing; austenitizing; tempering; postweld heat treatment, etc., shall be applied as required by the applicable Code requirements.

11.1.2 Heat treatment, other than pre-heat is performed by a subcontractor. The Quality Control Manager will either provide a company-generated procedure or review the subcontractor's procedure and verify the equipment calibration.

11.2 POSTWELD HEAT TREATMENT

11.2.1 All postweld heat treatment (PWHT) is performed as specified in a written procedure approved by the Quality Control Manager. At a minimum, the procedure will describe the placement of thermocouples, method of attachment and removal, heating and cooling rates, and the time at temperature.

11.3 OFF-SITE PWHT

11.3.1 All items that are postweld heat treated off-site will be appropriately marked by the Quality Control Inspector to maintain identification and traceability of the item.

11.3.2 The Quality Control Inspector shall inspect all items upon return for damage and identification. Nonconforming items will be controlled as described in Section 8 of this Quality Control Manual.

11.4 HEAT TREATMENT RECORDS

11.4.1 Postweld heat treatment records submitted by the subcontractor shall be a chart or record which records the conditions and items treated.

11.4.2 Heat treatment records are reviewed and approved by the Quality Control Manager.

11.4.3 Heat treatment records, including procedures, are submitted to the Authorized Inspector for review.

11.4.4 Heat treatment records are maintained in accordance with Section 13 of this Quality Control Manual.

CALIBRATION

12.1.1 All measurement, test, and examination equipment used for acceptance of pressure vessels shall be calibrated in accordance with this section.

12.1.2 The requirements of this section are not pertinent to measurement devices such as oven thermometers, rulers, tape measures, weld gages, fit-up gages, squares, and levels.

12.1.3 Calibration of measuring and testing equipment used by subcontractors shall be verified by the Quality Control Manager.

12.2 CALIBRATION PROCEDURES

12.2.1 The Quality Control Manager will assign a unique serial number to each item of measurement, test, and examination equipment which requires calibration. The serial number will be identified on the item or if the item is too small it shall be identified on the equipment container.

12.2.2 The Quality Control Manager is responsible for maintaining all equipment in calibration, unless out of service and clearly tagged or marked to prevent inadvertent use.

12.2.3 Calibration may be performed in-house or by a testing laboratory.

12.2.4 Pressure gages used for final pressure tests are checked for accuracy against a standard dead-weight tester or a calibrated master gage every six (6) months or when error is suspected. Gages shall be recalibrated or replaced at any time that there is reason to believe that they are in error. In-house standards, when used will be maintained in a protected environment and calibrated based on the frequency of calibration noted on the Calibration Log (Exhibit 9). They will also be calibrated when there is reason to believe that they are in error.

12.2.5 Linear measuring devices shall be calibrated annually. As an alternative, linear measuring devices may be verified against a calibrated standard prior to each use. If this method is chosen, the calibration frequency for the standard shall be indefinite unless deterioration or erosion is evident.

12.2.6 The frequency of calibration for other equipment is as required by the applicable Code, or if not addressed in the Code, as recommended by the equipment manufacturer or from experience with the equipment.

12.3 RECORDS

12.3.1 Each piece of calibrated equipment shall be listed on a Calibration Log maintained by the Quality Control Manager.

12.3.2 With the exception of calibration standard blocks, all calibrated equipment shall be identified through the use of a calibration decal attached to the equipment or the equipment container if too small for the decal to be attached. The decal shall list the equipment serial number, the date calibrated, the date calibration is due.

12.4 DISCREPANT EQUIPMENT

12.4.1 When equipment requiring calibration is found to be out of calibration it shall be removed from service by the Quality Control Manager.

12.4.2 All pressure vessels checked with equipment that is out of calibration shall be considered as nonconforming and controlled per Section 8 of this Quality Control Manual.

RECORDS RETENTION

13.1 All records in this Quality Control Manual are maintained by the Quality Control Manager in a storage file or in the applicable job file.

13.2 All specific job records are collected at the completion of the job by the Quality Control Manager. The Quality Control Manager performs the final review to assure the records are complete and correct prior to preparation and certification of the Manufacturer's Data Report.

13.3 The following records will be maintained, as applicable, for a minimum period of three years.

Manufacturer's Partial Data Reports
Manufacturing drawings
Design calculations, including any applicable Proof Test Reports
Material Test Reports and/or material certifications
Pressure parts documentation and certifications
Welding Procedure Specifications and Procedure Qualification Records
Welder/Welding Operator Performance Qualification Records for each welder who welded on the vessel
NDE interpretation reports
Repair procedure and records
Process control sheets
Heat treatment records and test results
Postweld heat treatment records
Nonconformances and dispositions
Pressure test records
Transfer forms
Weld Map (if used)

AUTHORIZED INSPECTOR

14.1 The Authorized Inspector (AI) is an individual holding a valid National Board Authorized Inspector Commission and designated as such by an Authorized Inspection Agency (AIA).

14.2 Savannah Tank & Equipment Corp. shall establish and maintain an inspection agreement with the AIA of Record. All required inspections are to be performed by the AIA of Record.

14.3 The Quality Control Manager is the company's liaison with the Authorized Inspection Agency and the Authorized Inspector.

14.4 The Quality Control Manager shall notify the National Board if there is a change of scope, ownership, name, location, address, or inspection agreement with the AIA.

14.5 The Authorized Inspector shall have access to all records and documents referenced in this Quality Control Manual.

14.6 A controlled copy of the Quality Control Manual is available to the Authorized Inspector at the shop or field site where Code activities are being carried out.

14.7 The Authorized Inspector and the Authorized Inspector Supervisor shall be provided with free access to the location where Code work is performed and to such parts of all plants and facilities that are concerned with the manufacture and supply of Code parts as requested.

14.8 The Quality Control Manager will provide any assistance requested by the Authorized Inspector during monitoring of the Quality Control Manual and to the Authorized Inspector Supervisor during ASME QAI-1/National Board audits.

REPAIRS AND ALTERATIONS

15.1 GENERAL

15.1.1 All shop and field repairs and alterations shall be carried out in accordance with the requirements of the National Board Inspection Code, the jurisdiction, and the applicable requirements of this Quality Control Manual. Authority and responsibility within Savannah Tank & Equipment Corp. shall remain as stated in Section 2 of this Quality Control Manual.

15.1.2 Repair and alteration activities requiring third party inspection shall be performed by a National Board Commissioned Inspector (Repair Inspector). The Repair Inspector shall be an individual who is currently employed by an Authorized Inspection Agency and either;

a. holds both an active National Board Inservice Inspector Commission and an R Endorsement, or
b. holds both an active National Board Authorized Inspector Commission and an R Endorsement.

15.1.3 The Quality Control Manager shall provide access to the Repair Inspector for all drawings, design calculations, specifications, procedures, Travelers, repair or alteration procedures, test results, and other documents as necessary to ensure compliance with the National Board Inspection Code.

15.1.4 The Quality Control Manager is responsible for compliance with this Quality Control Manual for repairs and alterations to pressure-retaining items in the shop and the field.

15.1.5 A REPAIR is defined as the work necessary to restore a pressure-retaining item to a safe and satisfactory operating condition.

15.1.6 ALTERATION is defined as any change in the item described on the original Manufacturer's Data Report which affects the pressure containing capability of the pressure-retaining item. Nonphysical changes such as an increase in the maximum allowable working pressure (internal or external), increase in design temperature, or a reduction in minimum temperature of a pressure-retaining item shall be considered an alteration.

15.1.7 RERATING – See definition of Alteration

15.2 ROUTINE REPAIRS

15.2.1 Routine Repairs as defined in the NBIC may be performed when acceptable to the jurisdiction and the Repair Inspector.

15.2.2 Prior to performing routine repairs the Quality Control Manager shall ensure that the routine repair is accepted by the Repair Inspector and the jurisdiction where the pressure-retaining item is installed. Subject to the acceptance of the jurisdiction, the Repair Inspector may give approval for routine repairs prior to the start of work.

15.2.3 Routine repairs under this organization shall be performed per this Quality Control System; however, the requirement for in-process involvement of the Repair Inspector and stamping may be waived if allowed by the jurisdiction and concurred with, by the Repair Inspector.

15.2.4 The Routine Repairs shall be documented on Form R-1 Report with a statement in the remarks section: "Routine Repair".

15.2.5 The Quality Control Manager, with the knowledge and understanding of jurisdictional requirements, shall be responsible for meeting jurisdictional requirements and the requirements of the NBIC.

15.2.6 Upon completion of the Routine Repair the Form R-1 Report and all supporting documentation shall be presented to the Repair Inspector for review and certification.

15.3 SCOPE OF WORK

15.3.1 The Engineering Manager shall determine the requirements that pertain to the specific construction code for the equipment being repaired or altered. When the standard governing the original construction is the ASME Code, repairs and alterations to pressure-retaining items shall conform, insofar as possible, to the section and edition of the ASME Code most applicable to the work planned. When the standard governing the original construction is not the ASME Code, repairs or alterations shall conform, insofar as possible to the edition of the construction standard or specification most applicable to the work. Where this is not possible or practicable, it is permissible to use other Codes, standards, or specifications, including the ASME Code, provided Savannah Tank & Equipment Corp. has the concurrence of the Repair Inspector and the jurisdiction where the pressure-retaining item is installed.

15.3.2 All work performed shall be in accordance with the applicable code of construction, NBIC, jurisdictional requirements, and this Quality Control Manual.

15.4 ORGANIZATION

15.4.1 The purpose of the Organization contained in Section 4 of this Quality Control Manual is to show the structural reporting of individuals responsible for the Quality Control System and levels of involved responsibility that assure the Quality Control System will meet the requirements of the NBIC.

15.5 DRAWINGS, DESIGN AND SPECIFICATIONS

15.5.1 Except as noted below, drawings, design and specification control shall meet the requirements of Section 5 of this Quality Control Manual.

15.5.2 A drawing as required by Section 5 of this Quality Control Manual is not always appropriate. If appropriate, a drawing shall be prepared to describe the repair or alteration. Drawings shall include sufficient information to satisfactorily perform the repair or alteration.

15.5.3 For alterations, calculations shall be completed prior to the start of any physical work.

15.5.4 For alterations to ASME Section I, Section IV, and Section VIII-1 2019 Edition and earlier, and B31.1, calculations, drawings and specifications shall be prepared (in-house, subcontracted or customer supplied), reviewed and approved by the Engineering Manager and QC Manager.

15.6 MATERIALS

15.6.1 Except as noted below, material control shall meet the requirements of Section 6 of this Quality Control Manual.

15.6.2 The Quality Control Manager shall verify the identification of existing materials and the identification of new material. The identification of the existing materials shall derive from original data, drawings, or pressure-retaining item records. If the existing material cannot be verified, the Quality Control Manager shall perform a chemical analysis and hardness testing, as a minimum of the unknown material to verify its weldability and strength or may elect to qualify a welding procedure.

15.6.3 Only material used for the original code of construction or acceptable alternates shall be used in making the repair or alteration. The materials used in making repairs or alterations shall conform insofar as possible to the original code of construction, construction standard or code selected, including the material specification requirements used for the work planned.

15.6.4 When a Manufacturer's Data Report is required by the original construction standard, a copy of the original Manufacturer's Data Report shall be obtained for use in the repair or alteration. When the original Manufacturer's Data Report cannot be obtained, agreements on the method of identifying the original material and establishing a basis for the repair or alteration shall be obtained from the Repair Inspector and the jurisdiction, when required.

15.6.5 Replacement parts shall be received, inspected, and tested in accordance with the requirements of the NBIC.

15.6.6 The QC Manager is responsible for verifying that no used replacement materials or Parts, that have previously been in service or considered as used material, is used in performing repairs and alterations.

15.7 METHOD OF PERFORMING WORK

15.7.1 The method of repairs or alterations to pressure-retaining items are subject to the acceptance of a Repair Inspector prior to the start of work. The Repair Inspector's authorization to perform a repair or alteration shall be obtained by the Quality Control Manager. The Repair Inspector shall determine that the method of repair or alteration is acceptable.

15.7.2 The Quality Control Manager will review the Traveler and other applicable documentation with the Repair Inspector for designation of hold points.

15.7.3 The Quality Control Manager shall apprise the Repair Inspector of work progress, and give notification in advance of approaching hold points as designated on the Traveler.

15.7.4 The Quality Control Manager shall arrange access to the pressure-retaining item for the Repair Inspector so that he may make any inspections deemed necessary to accept the repair or alteration.

15.8 CALIBRATION

15.8.1 Calibration control shall meet the requirements of Section 12 of this Quality Control Manual.

15.9 NONCONFORMITIES

15.9.1 Control of non-conformances shall meet the requirements of Section 8 of this Quality Control Manual.

15.10 WELDING, NONDESTRUCTIVE EXAMINATION AND HEAT TREATMENT

15.10.1 Except as noted below, Welding, Nondestructive Examination and Heat Treatment shall be in accordance with Sections 9, 10 and 11 of this Quality Control Manual respectively.

15.10.2 Standard Welding Procedures listed in the NBIC may be used as an alternative to Savannah Tank & Equipment Corp. Welding Procedure Specifications. The Quality Control Manager shall accept by certification, full responsibility for the application of the Standard Welding Procedure. When using Standard Welding Procedure Specifications all variables listed are considered essential and, therefore, deviations, modifications, amendments, or revisions are prohibited.

15.10.3 The following additional requirements will apply when repairing or altering a boiler using the ASME Code Section I 2019 or later edition:

a. Preheat maintenance for welding, including tack welding, shall be as defined by ASME Code Section I PW-38.
b. Welding Procedure Specifications shall specify preheat temperature in accordance with Table PW-38-1, and maximum interpass temperature in accordance with paragraph PW-38.9.
c. No welding or thermal cutting shall be performed when the base metal temperature is less than 50°F, or the temperature specified in the WPS, whichever is higher.

15.10.4 Under certain conditions, postweld heat treatment, in accordance with the original code of construction, may be inadvisable or impractical. In such instances, the alternative welding methods outlined in the NBIC are permitted. Competent technical advice shall be obtained from the manufacturer of the pressure-retaining item or from another qualified source. This advice will be especially necessary if abnormal service conditions exist as outlined in the NBIC.

15.10.5 Under certain conditions, nondestructive examination, in accordance with the original code of construction, may not be possible or practicable. In such instances, the alternative

nondestructive methods outlined in the NBIC acceptable to the Repair Inspector and the Jurisdiction, where required, may be used.

15.11 EXAMINATION AND TESTS

15.11.1 The Repair Inspector making the final acceptance inspection should be the Repair Inspector who authorized the repair or alteration, but in all cases, shall be employed by the same organization.

15.11.2 The integrity of repairs and alterations shall be verified by one, or a combination of the examination and test methods outlined in the NBIC.

15.11.3 Testing methods used shall be suitable for providing meaningful results to verify the integrity of the repair or alteration.

15.11.4 The examinations and test methods will be administered by the Shop/Field Supervisor as outlined on the Traveler.

15.11.5 The applications of examination and test methods shall meet the applicable requirements of Sections 7 and 10 of this Quality Control Manual.

15.11.6 For pressure testing of a field repair or alteration, the facility's in-house pressure gage may be used in lieu of a Savannah Tank & Equipment Corp.'s gage, if the facility's gage has been calibrated within the previous year. If a dial indicating pressure gage is used, it shall meet the requirements of paragraph 7.3.2 of this Quality Control Manual. If this method is chosen, the Quality Control Inspector shall obtain objective evidence of the calibration for that gage.

15.11.7 When a repair or alteration of a pressure-retaining item is completed, the Quality Control Inspector performs the final inspection and initials and dates the Traveler when all applicable Code requirements have been met.

15.12 The Repair Inspector shall make the final inspection and will authorize the stamping when satisfied that all applicable Code requirements have been met.

15.13 STAMPING

15.13.1 The Quality Control Manager retains custody and control of the "R" stamp in the shop and field.

15.13.2 When authorized by the Repair Inspector, the Quality Control Inspector shall apply the stamping or attach the nameplate adjacent to the original nameplate or stamping.

15.13.3 If necessary to remove the original Manufacturer's nameplate or stamping, the Repair Inspector will, subject to jurisdiction approval, witness the making of a rubbing or facsimile of the stamping, witness the obliteration or removal of the original stamping, witness transfer of the nameplate or stamping to a new location on the pressure-retaining item, and ensure its location is identified on the Form "R" Report. The Code Symbol/Certification Mark shall not be restamped unless allowed by the original code of construction.

15.13.4 The abbreviation 'ST & EC' may be stamped on the component or the nameplate as appropriate, in lieu of Savannah Tank & Equipment Corp..

15.13.5 The National Board "R" Symbol stamp may not be loaned out or used by anyone other than client name. The QCM shall return the "R" stamp to the National Board upon demand or if client name discontinues the use or if no inspection agreement exist with an AIA or if the Certificate of Authorization has expired a new certificate will not be issued.

15.14 CERTIFICATION, DISTRIBUTION AND RETENTION

15.14.1 The repair or alteration shall be documented on the current Form "R" Report. The Form "R" Report will be completed and certified by the Quality Control Manager. The final records and Form "R" Report are presented to the Repair Inspector for review and certification when satisfied all Code requirements have been met.

15.14.2 The Quality Control Manager shall distribute and maintain the Form "R" Report as described below.

DISTRIBUTION OF FORM R-1
Legible copies of completed Form R-1 Report, together with attachments, shall be distributed to the owner or user and Jurisdiction, if required, and shall be provided to the Repair Inspector and the inservice Authorized Inspection Agency of the pressure-retaining item upon request.

DISTRIBUTION OF FORM R-2
Legible copies of the completed Form R-2, together with attachments, shall be distributed to the owner-user, the "R" Certificate Holder responsible for the design, and the Jurisdiction, if required, and shall be provided to the Repair Inspector and the inservice Authorized Inspection Agency of the pressure-retaining item upon request.

15.15 RECORD RETENTION

15.15.1 The Quality Control Manager shall maintain a log or multiple logs documenting unique and sequentially numbered Form "R" reports that are registered with the National Board. The logs shall include, as a minimum; each form's unique registration number, type (R-1, R-2, etc.), description of work performed, date of acceptance by the Authorized Inspection Agency, and date the report was submitted to the National Board. The Quality Control Manager shall maintain all Form "R" Reports and all supporting records and documentation substantiating the summary of the work. These records, including Welders Logs, shall be easily retrievable and maintained in a storage file or the applicable job file for a minimum period of five years.

EXHIBITS
TABLE OF CONTENTS

EXHIBIT	NUMBER	REV	DATE
LIST OF MANUAL HOLDERS	1	0	04/14/2023
BILL OF MATERIAL	2	0	04/14/2023
PURCHASE REQUISITION	3A	0	04/14/2023
PURCHASE ORDER	3B	0	04/14/2023
RECEIVING INSPECTION REPORT	4	0	04/14/2023
MATERIAL TRACEABILITY LOG	4A	0	04/14/2023
TRAVELER	5	0	04/14/2023
WELD MAP	6	0	04/14/2023
WELDERS LOG	7	0	04/14/2023
NON-CONFORMANCE REPORT	8	0	04/14/2023
CALIBRATION LOG	9	0	04/14/2023
DESIGN CONTINUITY LOG	10	0	04/14/2023

SAVANNAH TANK & EQUIPMENT CORP.

LIST OF MANUAL HOLDERS EDITION:____

NAME	TITLE	MANUAL NO:	REV LEVEL	DATE

BILL OF MATERIAL

BILL OF MATERIAL					
JOB NUMBER:					
QTY	UNIT	DESCRIPTION/SPEC	VENDOR	PO NUMBER	MATERIAL ID

PURCHASE REQUISITION

Savannah Tank & Equipment Corporation

BILL OF MATERIAL 1517 Telfair Road Savannah, Georgia 31415

DATE:	5/17/2017	FINAL		REL. FOR PREL.	5/17/2017	STOCK CHECK COMP.
JOB NUMBER:	28/17			PREL. COMP.	5/17/2017	FINAL COMPLETED
DESCRIPTION:	VESSEL			REV. PRELIMINARY	5/17/2017	REL FOR FINAL
EQUIP. NUMBER:			5/17/2017	REL FOR STOCK CHK	5/22/2017	REVISED FINAL
		PAGE 1 OF 1		RET FOR ST. CHK		AS BUILT

ITEM	QTY	DESCRIPTION	MATERIAL SPECIFICATION	VENDOR	PO #/DELIVERY	NOTES:
		ALL DIMENSIONS ARE APPROXIMATELY USE DRAWING FOR CUTS				
CAPS	2	PIPE CAPS, 8" STD	SA234-WPW B16.9	FERGUSON	31894 5-23-17	
SHELL	1	PIPE, 8" SCH STD X 12"	SA53B ERW	FERGUSON	31894 5-23-17	
HALF CPLG - A, B	2	HALF THRD. CPLG, 1/4" 3000#	SA105 B16.11	FERGUSON	31894 5-23-17	
PLUG - A, B	2	SQUARE HEAD PLUG, 1/4" 3000#	SA105 B16.11	FERGUSON	31894 5-23-17	
HALF CPLG - C	1	HALF THRD. CPLG, 3/4" 3000#	SA105 B16.11	FERGUSON	31894 5-23-17	
PLUG - C	1	SQUARE HEAD PLUG, 3/4" 3000#	SA105 B16.11	FERGUSON	31894 5-23-17	
HALF CPLG - D	1	HALF THRD. CPLG, 3/4" 3000#	SA105 B16.11	MOORE	31906 5-23-17	
PLUG - D	1	HEX. HEAD PLUG, 3/4" 3000#	SA105 B16.11	MOORE	31906 5-23-17	
LIFT LUG	1	PLATE, 1/4" X 3" X 4" SHAPED W/ HOLE	SA36	STOCK	HTR 1500614-5	$2
SUPPORT	2	PLATE, 1/4" X 10" X 10", CUT TO SUIT	SA36	STOCK	HTR 1500614-5	$5
N'PLATE BRKT	1	SHEET, 12 GAUGE 4 3/4" X 15" (FORM)	SA240-316/316L	STOCK	HTR 1500485	$5 PRODUCTION OFFICE
NAMEPLATE	1	NAMEPLATE	ALUMINUM	STOCK		$10 PRODUCTION OFFICE

REQUISITIONED BY: NICK PATEL APPROVED BY:

PURCHASE ORDER

SAVANNAH TANK
AND EQUIPMENT CORPORATION

Savannah Tank and Equipment Corp.
1517 Telfair Road
Savannah, GA 31415
United States of America

Ph: 912-233-8902 Fax: 912-233-8921

Purchase Order
Number: 31906 Date: 22-May-17

To	Ship To
MOORE INDUSTRIAL SUPPLY P.O. BOX 2427 SAVANNAH, GA 31402 United States of America	Savannah Tank and Equipment Corp. 1517 Telfair Road Savannah, GA 31415 United States of America
Ph: Fax:	Ph: 912-233-8902 Fax: 912-233-8921

Terms	Ship Via	FOB	Issued By
Net 30 Days		Destination	NICK

Quantity	Description	Unit Price	Amount
1 ea Line: 001	CPLG SA105 3/4 3000 HALF THRD Coupling 105 3/4"3000 Half Th SA105 B16.11 Certifications Required Qty: 1 Job: 17-028 Due: 23-May-17	$1.15 ea	$1.15
1 ea Line: 002	HEX. PLUG SA105 3/4 3000 Hex. Plug 105 3/4" 3000# SA105 B16.11 Certifications Required Qty: 1 Job: 17-028 Due: 23-May-17	$0.84 ea	$0.84
	Savannah Tank accepts NO material from China excluding nuts & bolts. Our purchase order number must appear on all invoices, shipping papers and packages. Invoices can be submitted electronically to: APInvoice@savannahtank.com Material test reports are to be furnished with shipment. All Materials must be delivered free from Defects in Material and Workmanship. Vendor will repair or replace defective parts at the vendor's expense for a period of eighteen(18) months from the date material is received at Savannah Tank. Materials must be protected from damage in transit. Plate items to be palletized can not exceed 5,000 pounds. Unit cost must reflect the full cost of the item. Any additional charges (such as Environmental Fees, Fuel Surcharges, and Processing Fees) that appear on the invoice that were not reflected in your response to the RFQ and/or included in our PO will not be paid. Deliveries to be made on Monday - Thursday between 8:30 AM and 1:30 PM. Material to be marked in accordance with the identification requirements of ASME Section VIII & II, Division 1. Heads shall be in accordance w/ paragraphs UG-79 & UG-81 of Sect. VIII, Div. 1 ASME B & PV code, latest edition.	Total:	$1.99 US Dollars

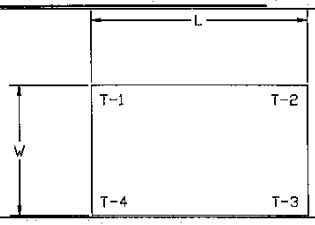
Page 1 of 1

RECEIVING INSPECTION REPORT

Job #:_____ PO#_____ Date: _____ M&TE Serial # Used_____

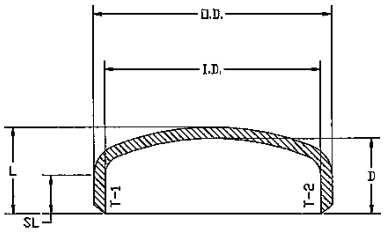
Plate **Quantity** _____. **Coded Marking** _____

Spec. No.:	Grade:	Heat#:	
T-1:	T-2:	T-3:	T-4:
W:	L:	MTR Received: Yes ☐ No ☐	
Original Identification Markings:			
Accepted:		Rejected:	



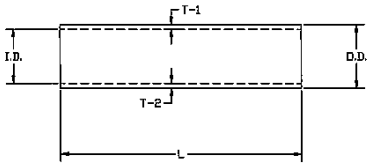
Heads & Caps **Quantity** _____. **Coded Marking** _____

Spec. No.:	Grade:	Heat#:	
Stamping:			
O.D.:	I.D.:	D:	L:
SL:	T-1:	T-2:	MTR Received: Yes ☐ No ☐
Original Identification Markings:			
Accepted:		Rejected:	



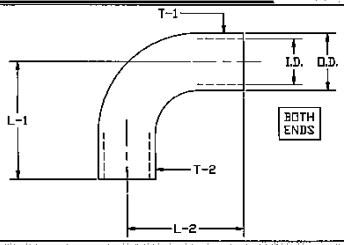
Pipe **Quantity** _____. **Coded Marking** _____

Spec. No.:	Grade:	Heat#:	
I.D.:	O.D.:	T-1:	T-2:
L:	MTR Received: Yes ☐ No ☐		
Original Identification Markings:			
Accepted:		Rejected:	



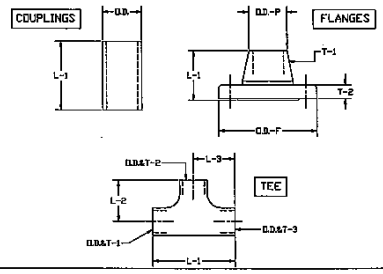
Ells **Quantity** _____. **Coded Marking** _____

Spec. No.:	Grade:	Heat#:	
Stamping:		MTR Received: Yes ☐ No ☐	
I.D.#1:	I.D.#2:	O.D.#1:	O.D.#2:
T-1:	T-2:	L-1:	L-2:
Original Identification Markings:			
Accepted:		Rejected:	



Other: **Quantity** _____. **Coded Marking** _____

Spec. No.:	Grade:	Heat#:
Description:		
Stamping:	MTR Received: Yes ☐ No ☐	
List Inspection Points:		
Original Identification Markings:		
Accepted:	Rejected:	



Material Accepted _____ Date _____

MATERIAL TRACEABILITY LOG

PRODUCT	TRACE CODE	DATE ASSIGNED	PO NO.	ORIGINAL PRODUCT MARKING	MTR / CofC	
					YES	NO
					☐	☐
					☐	☐
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					☐	☐

Savannah Tank & Equipment Corp.

TRAVELER

SAV TANK# 028-17
NB#
ISSUED BY: JLF
DATE: 5/22/2017

NOTE: FABRICATION MAY NOT PROCEED PAST A HOLDPOINT UNTIL INSPECTED & SIGNED OFF.

	Q.C. HP	A.I. HP	Q.C. INT	DATE	A.I. INT	DATE
INITIAL TRAVELER REVIEW, REVIEW DRAWINGS,	H	H				
CALCS, DETAILS, SPECIFICATIONS	H	H				
PRE-FAB VERIFICATION						
WELDER QUALIFICATIONS	H					
THICKNESS VERIFICATION	H					
MILL TEST REPORTS	H					
HEAT NUMBER TRANSFER	H					
FIT-UP INSPECTION						
HEAD TO SHELL - TOP	H					
HEAD TO SHELL - BOTTOM	H					
SHELL TO SHELL	H					
MANWAY(S)						
NOZZLE(S)	H					
JACKET TO HEAD						
JACKET TO SHELL						
OTHER _____						
A.I. HOLDPOINT _____	H	H				
A.I. HOLDPOINT (INTERNAL INSPECTION)	H	H				
WELD MAP	H	H				
NDE PROCEDURES						
STRESS RELIEF _____						
REVIEW RADIOGRAPHS						
DYE PENETRANT						
VISUAL	H	H				
HYDROSTATIC TEST — GAUGE #						
VESSEL 910 PSIG 22	H	H				
JACKET - PSIG						
INTERNAL COIL - PSIG						
FINAL AND/OR INTERNAL INSPECTION	H	H				
VERIFY AND APPLY CODE STAMP	H	H				
SIGN ASME DATA REPORT	H	H				
NCR REPORTS						
MISCELLANEOUS						

WELD MAP

Savannah Tank & Equipment Corp.

EQUIP # - SAV TANK # 028-17 PREPARED BY: _____

DESCRIPTION: TEST GAUGE VESSEL DATE: _____

VESSEL WELD MAP

WELD PROC. GT-10, FC-38 & FC-39

SEAMS	I/S WELD	O/S WELD	WPS#
C1	-		10
C2	-		10

NOZZLE	FLG-PIPE	WPS#	INSTALL	REPAD	WPS#
A	-	-		-	10
B	-	-		-	10
C	-	-		-	10
D	-	-		-	10

OTHER WELDS-EXT		WPS#
SUPPORTS		
SUPPORT /TANK		38
SUPPORT/REPAD		
LIFT LUG		
LIFT LUG /TANK		38
LIFT LUG/REPAD		
INS RINGS		
VAC RINGS		
GROUND		
LADDER		
LADDER TANK CLIPS		
HANDRAIL		
HANDRAIL TANK CLIPS		
PLATFORM		

OTHER WELDS-EXT		WPS#
PLATFORM SUPP/TANK		
REST PLATFORM		
REST PLTFRM SUPP/TANK		
JACKET		
NAMEPLATE BRACKET		39
OTHER WELDS-INT		WPS#
BAFFLE		
BAFFLE SUPP		
COIL		
COIL SUPP		
WEAR PLATE		

GM/FC QUALIFICATIONS

NAME	EMP#	GM-01 FC-54	GM-02 FC-57	GM-03 FC-58	GM-04 FC-59	GM-05 GM-60	GM-06	FC-14	FC-38	FC-38HT	FC-39	FC-44	FC-45	FC-46	FC QUAL
BRASILE, ANDREW	340	7/15/2016	8/8/2016					3/7/2017	2/19/2017	7/15/2016	3/30/2017	6/7/2016	6/7/2016	6/7/2016	8/30/2017
BUSH, JAMES	350							3/7/2017	5/4/2017	10/14/2016	10/14/2016	10/14/2016	10/14/2016	10/14/2016	10/4/2017
CHAVIS, SAMMY	103	3/6/1998 3/12/2014	11/14/2016 7/8/2015	4/11/1997 4/23/2014	9/28/2007	4/11/1997	4/11/1997	8/10/2016	5/4/2017	10/24/2011	3/28/2011	7/8/2015	7/8/2015	7/8/2015	10/4/2017
DELOACH, BRIAN	353							3/30/2017	4/20/2017	11/29/2016	11/29/2016	11/29/2016	11/29/2016	11/29/2016	9/20/2017
GARRISON, ERIC	365								4/11/2017	4/11/2017					9/11/2017
LAWRENCE, THURLTO	327	7/12/2016	11/28/2016					3/30/2017	5/4/2017	5/5/2016	3/1/2016	3/1/2016	3/1/2016	3/1/2016	10/4/2017
LOVE, RAYMOND	356							3/30/2017	5/4/2017	3/1/2017	3/1/2017	3/1/2017	3/1/2017	3/1/2017	10/4/2017
McMOORE, KEVIN	357							3/30/2017	3/1/2017	3/1/2017	3/1/2017	3/1/2017	3/1/2017	3/1/2017	8/30/2017
SELLARS, MAURICE	317	9/12/2016		5/20/2016 9/12/2016	5/17/2016		5/17/2016	3/30/2017	4/20/2017	5/21/2016	1/4/2016	1/4/2016	1/4/2016	1/4/2016	9/20/2017
SHUMAN, CORY	284	5/22/2014		3/11/2014		5/24/2016		3/30/2017	4/20/2017	10/29/2012	5/24/2016	11/14/2016	7/28/2015	7/8/2015	9/20/2017
SHUMAN, RANDALL	338							3/30/2017	4/20/2017	9/21/2016	6/7/2016	6/7/2016	6/7/2016	6/7/2016	9/20/2017
SHUMAN, SCOTT	130	7/12/2016 9/12/2016	6/16/2016	9/12/2016				3/31/2016	4/20/2017	10/24/2011	9/3/2014	1/23/1998	1/23/1998	1/23/1998	9/20/2017
SICKLES, CLINTON	375							5/4/2017	5/4/2017	5/4/2017	5/4/2017	5/4/2017	5/4/2017	5/4/2017	10/4/2017
THOMPSON, PAUL	352							2/19/2017	11/14/2016	11/14/2016	11/14/2016	11/14/2016	11/14/2016	11/14/2016	7/19/2017
WELLS, TRAVIS	273	6/16/2016 3/12/2014	10/9/2012	3/12/2014				2/16/2017	5/4/2017	10/24/2011	4/9/2014	4/4/2016	9/15/2010	9/15/2010	10/4/2017

SAVANNAH TANK & EQUIPMENT CORP

NONCONFORMANCE REPORT

NCR No.		DATE:	
JOB No.		OTHER:	
ITEM DESCRIPTION:			
DESCRIPTION OF NONCONFORMANCE:			
QC INSPECTOR:			
RECOMMENDED DISPOSITION			
☐ USE-AS-IS	☐ REPAIR	☐ RETURN TO VENDOR	☐ OTHER
QC INSPECTOR:		DATE:	
DISPOSITION APPROVAL/ACCEPTANCE			
ENGINEERING MGR APPROVAL:		DATE:	
QUALITY CONTROL MGR APPROVAL:		DATE:	
AUTHORIZED INSPECTOR ACCEPTANCE:		DATE:	
DISPOSITION COMPLETED			
COMPLETED DISPOSITION AS RECOMMENDED? Yes ☐ No ☐ COMMENTS:			
QUALITY CONTROL MGR APPROVAL:		DATE:	
AUTHORIZED INSPECTOR ACCEPTANCE:		DATE:	

CALIBRATION LOG

EQUIPMENT DESCRIPTION	SERIAL NO.	DATE CALIBRATED	NEXT DUE DATE	COMMENTS

DESIGN CONTINUITY LOG

NAME	DESIGNATED		Designation Date
	Certifying Engineer ☐ Engineer ☐ Designer ☐		
List Section VIII-1 design work – Date and Job No.			